# Hydrapress HP-200

## Hydraulic Press: Operation and Maintenance Manual

Document HP200-OM-04. Sample manual for the YANTRA prototype. The machine and all part numbers are fictional.

# 1. Safety

Read this section before operating the press. Only trained operators may use the machine.

1.1 Always keep both hands on the two start buttons during the pressing stroke. Never reach into the die area while the motor is running.

1.2 The emergency stop button (red, mushroom head) stops the motor and removes control power. Release it by twisting clockwise.

1.3 The front guard door is fitted with a safety interlock. Never bypass, tape over or adjust the interlock switch.

1.4 Hydraulic oil under pressure can pierce skin. Before opening any fitting, stop the motor and wait 2 minutes for the system to depressurise.

1.5 Hot oil and the oil cooler can cause burns. Allow the machine to cool before servicing.

1.6 Repairs marked 'qualified technician' in this manual must not be attempted by operators.

# 2. Technical data

| Item | Value |
|---|---|
| Maximum pressing force | 200 kN |
| Maximum system pressure | 160 bar |
| Motor | 7.5 kW, 400 V, 3 phase |
| Hydraulic oil | ISO VG 46, 40 litres |
| Normal oil temperature | 40 to 55 °C |
| Stroke time (empty) | about 8 seconds |

# 3. Routine maintenance

Daily: check the oil level in the sight glass and look for leaks around the cylinder and hoses.

Weekly: check the return filter indicator and clean the oil cooler fins.

Every 500 hours: replace the return filter element. Every 2000 hours: change the hydraulic oil.

Every year: have the guard door interlock tested by a qualified technician.

# 4. Troubleshooting

Find the fault in the list below. Each fault shows the symptoms, then the possible causes with a check to identify the cause and the remedy steps. Carry out the steps in order.

## Fault 4.1: Ram does not move

Symptoms: ram does not move when the cycle is started; press will not go down; motor is running but the ram stays still; ram not moving down or up; hydraulic cylinder not responding.

### *Possible cause 1.1: Motor is not starting: emergency stop is engaged or the motor overload relay has tripped.*

How to check: The motor is silent and the motor fan is not turning.

Remedy:

1. Check that the red emergency stop button is released: twist it clockwise until it pops out.
2. Press the green MOTOR START button once.
3. If the motor still does not start, press the external overload reset knob on the cabinet door. Do not open the cabinet.
4. Press MOTOR START again and confirm the motor runs.

### *Possible cause 1.2: Hydraulic oil level is too low.*

How to check: The motor is running, the gauge reads zero, and the sight glass shows oil below the MIN mark.

Remedy:

1. WARNING: Stop the motor with the red STOP button before opening the filler cap.
2. Read the oil sight glass on the reservoir. The level must be between the MIN and MAX marks.
3. Remove the filler cap and top up with hydraulic oil ISO VG 46 until the level reaches the MAX mark.
4. Refit the filler cap and start the motor.
5. Run three empty cycles and check the sight glass again. Top up if the level fell below MAX.

### *Possible cause 1.3: Directional valve solenoid Y1 is not energised (blown fuse F4 or loose connector).*

How to check: The motor runs and the gauge shows pressure, but the ram does not move and the green LED on valve Y1 stays off.

Remedy:

1. WARNING: Press the emergency stop before touching any connector or fuse.
2. Check that the plug on valve Y1 is fully seated and its locking ring is tight.
3. Open fuse holder F4 on the control panel. Replace the fuse with a 2 A glass fuse if the wire inside is broken.
4. Release the emergency stop, start the motor and start a cycle.
5. Confirm the green LED on valve Y1 lights up and the ram moves.

## Fault 4.2: Ram moves slowly

Symptoms: ram is moving very slowly; press cycle takes much longer than usual; slow pressing stroke; ram speed has dropped.

### *Possible cause 2.1: Return line filter is clogged.*

How to check: The filter indicator is in the red zone.

Remedy:

1. WARNING: Stop the motor and wait 2 minutes for the pressure to drop to zero before opening the filter housing.

2. Unscrew the filter bowl and remove the old element.

3. Fit a new element, part number HF-200-R, and tighten the bowl by hand plus a quarter turn.

4. Start the motor and check that the indicator returns to green.

5. Wipe away any spilled oil around the filter.

### *Possible cause 2.2: Oil is overheating because the cooler fan is not running or the cooler is blocked.*

How to check: The thermometer reads above 55 °C and the filter indicator is green.

Remedy:

1. WARNING: Hot oil can cause burns. Let the press cool for 15 minutes before touching the cooler.

2. Check that the cooler fan spins when the oil is above 50 °C. If it does not, check fan fuse F5.

3. Brush dust and lint off the cooler fins with a soft brush.

4. Restart the press and watch the thermometer for 10 minutes. It must stay at or below 55 °C.

### *Possible cause 2.3: Flow control valve FC1 has been turned down.*

How to check: The oil temperature is normal, the filter indicator is green, and the ram has been slow since the first cycle of the shift.

Remedy:

1. Locate flow control valve FC1 beside the main cylinder.

2. Loosen the lock nut on the FC1 adjuster.

3. Turn the adjuster anticlockwise one turn at a time, running one cycle after each turn, until the stroke takes about 8 seconds.

4. Tighten the lock nut and mark the adjuster with a paint pen.

## Fault 4.3: Pressing force too low (pressure does not reach the set value)

Symptoms: pressure gauge does not reach the set pressure; press is weak and cannot form the part; low pressing force; system pressure is low.

### *Possible cause 3.1: Relief valve RV1 is set too low.*

How to check: The pressure rises when RV1 is turned clockwise, the ram holds its position, and the oil looks normal.

Remedy:

1. Loosen the lock nut on relief valve RV1.

2. Turn the adjuster clockwise in quarter turns while watching the gauge.

3. WARNING: Never set the pressure above 160 bar.

4. Tighten the lock nut when the gauge reads 160 bar during a pressing stroke.

### *Possible cause 3.2: Air in the hydraulic system.*

How to check: The oil in the reservoir is foamy or milky and the pump is noisier than normal.

Remedy:

1. WARNING: Stop the motor before touching any fitting.

2. Check that the oil level is above the MIN mark and top up if needed.

3. Check that the pump suction hose clamps are tight (hand tight plus a quarter turn).

4. Start the motor at low pressure and open the bleed screw at the top of the cylinder half a turn. Close it when oil comes out without bubbles.

5. Run five empty cycles. The foam should clear within 10 minutes.

### *Possible cause 3.3: Piston seal inside the main cylinder is leaking.*

How to check: The ram creeps downward when held under load, and turning RV1 has no effect.

Qualified technician required. The cylinder must be dismantled to replace the piston seal. Do not attempt this repair yourself.

Make the machine safe:

1. Stop the press and press the emergency stop.

2. Switch off the main isolator and apply your lock-out padlock.

### *Possible cause 3.4: Relief valve RV1 is stuck partly open because of dirt.*

How to check: The ram holds its position and the oil is clean, but turning RV1 clockwise changes nothing.

Remedy:

1. WARNING: Stop the motor and wait for the pressure to drop to zero before removing RV1.

2. Note the number of adjuster turns, then unscrew the RV1 cartridge with a 24 mm spanner.

3. Rinse the cartridge in clean hydraulic oil. Do not use rags or compressed air.

4. Refit the cartridge and turn the adjuster back to the noted position.

5. Start the motor, set the pressure to 160 bar on the gauge and tighten the lock nut.

## Fault 4.4: Hydraulic oil leak

Symptoms: oil is leaking from the press; puddle of oil under the machine; oil dripping from the cylinder or hoses; wet oil patches on the press.

### *Possible cause 4.1: Rod seal on the main cylinder is worn.*

How to check: Oil runs down the cylinder rod and collects at the cylinder gland.

Qualified technician required. Replacing the rod seal means removing the cylinder from the frame. Do not attempt this repair yourself.

Make the machine safe:

1. Stop the press and press the emergency stop.

2. Place a drip tray under the cylinder and keep people away from the area.

### *Possible cause 4.2: A hose or fitting is loose or damaged.*

How to check: A hose or its end fitting is wet with oil, or the hose is cracked, swollen or chafed.

Remedy:

1. WARNING: Stop the motor and wait 2 minutes for the pressure to drop to zero.

2. Wipe the area dry and find the exact point where the oil comes out.

3. If a fitting is wet, tighten it with a 22 mm spanner to 35 Nm.

4. If the hose itself is cracked, swollen or chafed, stop work and fit a new hose, part number HP-200-H3.

5. Run three cycles and check that the fitting stays dry.

### *Possible cause 4.3: Reservoir filler cap or tank cover gasket is loose.*

How to check: Oil seeps from the top of the reservoir around the filler cap or the tank cover.

Remedy:

1. WARNING: Stop the motor before touching the reservoir.

2. Tighten the filler cap by hand.

3. Tighten the 8 tank cover screws in a cross pattern to 10 Nm.

4. Wipe up the oil and check again after three cycles.

## Fault 4.5: Cycle will not start (start buttons have no effect)

Symptoms: cycle start button does nothing; press does not start when I press both buttons; machine will not start a cycle; start buttons are not responding.

### *Possible cause 5.1: Control fuse F2 is blown.*

How to check: The Control Power lamp is off.

Remedy:

1. WARNING: Press the emergency stop and switch off the main isolator before opening the fuse holder.

2. Open fuse holder F2 on the side of the panel and look at the 4 A fuse. Replace it if the wire inside is broken.

3. Close the fuse holder, switch the isolator on and release the emergency stop.

4. If the new fuse blows straight away, stop and call a technician.

### *Possible cause 5.2: The guard door is open or not latched.*

How to check: Control Power is lit, the Safety Circuit lamp is lit, and the guard door is open.

Remedy:

1. Close the front guard door fully until it clicks.

2. Check that nothing, such as a tool or a part, is trapped in the door edge.

3. Confirm the Safety Circuit lamp goes off.

4. Press both start buttons together to start the cycle.

### *Possible cause 5.3: Guard door switch S1 is misaligned or faulty.*

How to check: Control Power is lit, the Safety Circuit lamp is lit, and the guard door is fully closed.

Qualified technician required. Safety interlocks must only be adjusted or replaced by a qualified technician and must never be bypassed. Do not attempt this repair yourself.

Make the machine safe:

1. Stop work and leave the guard door closed.

2. Do not bypass or tape over the door switch.

### *Possible cause 5.4: The two start buttons are not pressed together.*

How to check: Control Power is lit, the Safety Circuit lamp is off, and nothing happens when the buttons are pressed one after the other.

Remedy:

1. Press and hold both green start buttons at the same time. They must be pressed within 0.5 seconds of each other.

2. Check that neither button is stuck in. A stuck button must spring back when released.

3. Wipe oil and dirt off both buttons with a dry cloth, then try again.

## Fault 4.6: Hydraulic pump is noisy

Symptoms: hydraulic pump is making a loud noise; whining sound from the pump; knocking or grinding noise from the pump unit; pump is very noisy.

### *Possible cause 6.1: Oil level is too low, so the pump sucks in air.*

How to check: The sight glass shows oil below the MIN mark.

Remedy:

1. WARNING: Stop the motor with the red STOP button before opening the filler cap.

2. Remove the filler cap and top up with hydraulic oil ISO VG 46 until the level reaches the MAX mark.

3. Refit the filler cap, start the motor and listen. The noise should stop within a minute.

### *Possible cause 6.2: The pump suction strainer is blocked.*

How to check: The oil level is normal and the pump inlet vacuum gauge is in the red zone.

Remedy:

1. WARNING: Stop the motor and close suction ball valve V1 before removing the strainer cover.

2. Remove the strainer cover and pull out the mesh basket.

3. Rinse the basket in clean hydraulic oil and refit it.

4. Open V1 fully, restart the motor and confirm the vacuum gauge reads in the green zone.

### *Possible cause 6.3: Pump bearings are worn.*

How to check: The oil level is normal, the vacuum gauge is in the green zone, and the pump makes a steady grinding or knocking noise.

Qualified technician required. The pump must be removed from the press and rebuilt. Do not attempt this repair yourself.

Make the machine safe:

1. Stop the press and press the emergency stop.

2. Switch off the main isolator and apply your lock-out padlock.